

1.1

$(i, j) \in \Omega$, where $i, j \in \{1, 2, 3, 4, 5, 6\}$.

Let $A = \{\text{Second Roll is Higher than the first}\}$

First Roll	P(A)
1	5/6
2	4/6
3	3/6
4	2/6
5	1/6
6	0/6

Thus $|A| = 5 + 4 + 3 + 2 + 1 = 15$

$|\Omega| = 6 * 6 = 36$

$P(A) = \frac{|A|}{|\Omega|} = \frac{15}{36} = \frac{5}{12}$

1.5

a

$\Omega = (a_1, a_2, a_3, a_4, a_5) | (a_1, a_2, a_3, a_4, a_5 \in \{1, 2, 3, 4, \dots, 40\}) \wedge \forall i, j (i \neq j \rightarrow a_i \neq a_j)$

$|\Omega| = {}_{40}P_5 = \frac{40!}{35!}$

$\forall \omega \in \Omega P(\{\omega\}) = 35!/40! = \frac{1}{40*39*38*37*36} = \frac{1}{78960960}$

b

We can think of this problem as choosing 3 numbers from the set of the even numbers 1-40 and then choosing 2 numbers from the set of odd numbers 1-40.

This results in the combinations of lottery numbers being $\frac{20!}{17!3!} \frac{20!}{18!2!}$

If we want to find the permutations of these then we multiply by 5! so this results in $\frac{5!20!20!}{3!2!17!18!} = 25992000$ different permutations of lottery numbers with exactly 3 evens.

Thus to find the probability we divide by $|\Omega|$ which results in $P(3E) = \frac{5!20!20!35!}{3!2!17!18!40!} = \frac{25992000}{78960960} = 0.3291753292$

1.7

a

$$P(GYG) = \frac{3}{7} \frac{4}{6} \frac{2}{5} = \frac{4}{35}$$

b

$3P(GYG) = 12/35$ this is because the probability of any order of 2 green and one yellow is the same.

1.10

a

$$\Omega = \{\infty, 1, 2, 3, \dots\}$$

$$\forall \omega \in \Omega \ P(\omega) = \left(\frac{5}{6}\right)^{\omega-1} * \frac{1}{6}$$

b

$$P(\omega = \infty) = \lim_{\omega \rightarrow \infty} \left(\frac{5}{6}\right)^{\omega-1} * \frac{1}{6} = 0$$

$$P(\{\infty\}) = 0$$

1.12

a

$$P(\omega \leq 3) = P(1) + P(2) + P(3) = \frac{1}{6} + \frac{5}{36} + \frac{25}{216} = \frac{91}{216}$$

It is also important to note that $P(1)$, $P(2)$, $P(3)$ are disjoint

b

$$P(E\omega) = 5/36 + 125/(216 * 6) + \dots$$

$$\text{Using geometric series sum: } P = \frac{5/36}{1-(5/6)^2} = \frac{5}{11}$$

1.20

a

$$\Omega = \{1, 2, 3, 4, 5, 6\}^4$$

$$\forall \omega \in \Omega P(\{\omega\}) = \frac{1}{|\Omega|} = \frac{1}{6^4}$$

b

$$|\Omega| = 6^4$$

$$B = \{\text{No 5's}\} \cup \{\text{Only 1 5}\}$$

$$|\{\text{No 5's}\}| = 5^4$$

$$|\{\text{Only 1 5}\}| = 4 * 5^3 \text{ (because the 5 can be in 4 different positions and other numbers are 5 options \{non 5\})}$$

$$|B| = 5^4 + 4 * 5^3$$

$$P(B) = \frac{5^4 + 4 * 5^3}{6^4} = \frac{1125}{6^4}$$

$$A = \{\text{Only 5's}\} \cup \{\text{One non 5}\} \cup \{\text{2 5's}\}$$

$$|\{\text{Only 5's}\}| = 1$$

$$|\{\text{Only one non 5}\}| = 4 * 5 = 20 \text{ (non 5 can be in 4 different places with 5 different values)}$$

For the set of 2 5's, there are 6 different orientations of the numbers where $x, y \in \{1, 2, 3, 4, 6\}$

$$xy55 \ x5y5 \ x55y \ 5xy5 \ 5x5y \ 55xy$$

$$\text{Thus } |\{2 \text{ 5's}\}| = 6 * 5^2 = 150$$

$$\text{Therefore } P(A) = \frac{150+20+1}{6^4} = \frac{171}{6^4}$$

c

$$A \cup B = \Omega$$

$$P(A) + P(B) = 1$$

$$\rightarrow \frac{171}{6^4} + \frac{1125}{6^4} = 1$$

$$\rightarrow \frac{1296}{6^4} = 1$$

$$\rightarrow \frac{1296}{1296} = 1$$

So the answer to part b holds true

1.22

a

$$\Omega = (i, j) | i \in \{\text{hearts, clubs, spades, diamonds}\}, j \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A\}$$

$$\forall \omega \in \Omega \ P(\omega) = \frac{1}{|\Omega|} = \frac{1}{4 \cdot 13} = \frac{1}{52}$$

b

What is the probability of picking an Ace from the deck that is not the Ace of Spades?

c

$$\text{Suppose } \exists A \text{ s.t. } P(A) = \frac{1}{5}$$

$$\text{By fact 1.8, then } P(A) = \frac{|A|}{|\Omega|} = \frac{1}{5}$$

$$\text{Plugging in our known values: } \frac{|A|}{52} = \frac{1}{5}$$

$$\text{When solving for } |A|: |A| = \frac{52}{5} = 10.4$$

This is not possible because the size of a set cannot be a non whole number. Therefore $\nexists A$ s.t. $P(A) = \frac{1}{5}$

1.34

$$\Omega = (i, j) \forall i, j (0 \leq i \leq 1, 0 \leq j \leq 1)$$

Circle lies within the square iff $(\frac{1}{3} < i < \frac{2}{3}) \wedge (\frac{1}{3} < j < \frac{2}{3})$

The range $(\frac{1}{3}, \frac{2}{3})$ is one third the size of the possible range. Thus the probability the i is within the range is $\frac{1}{3}$ and same for j .

Since we apply this range to both i and j the resulting probability is $\frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$